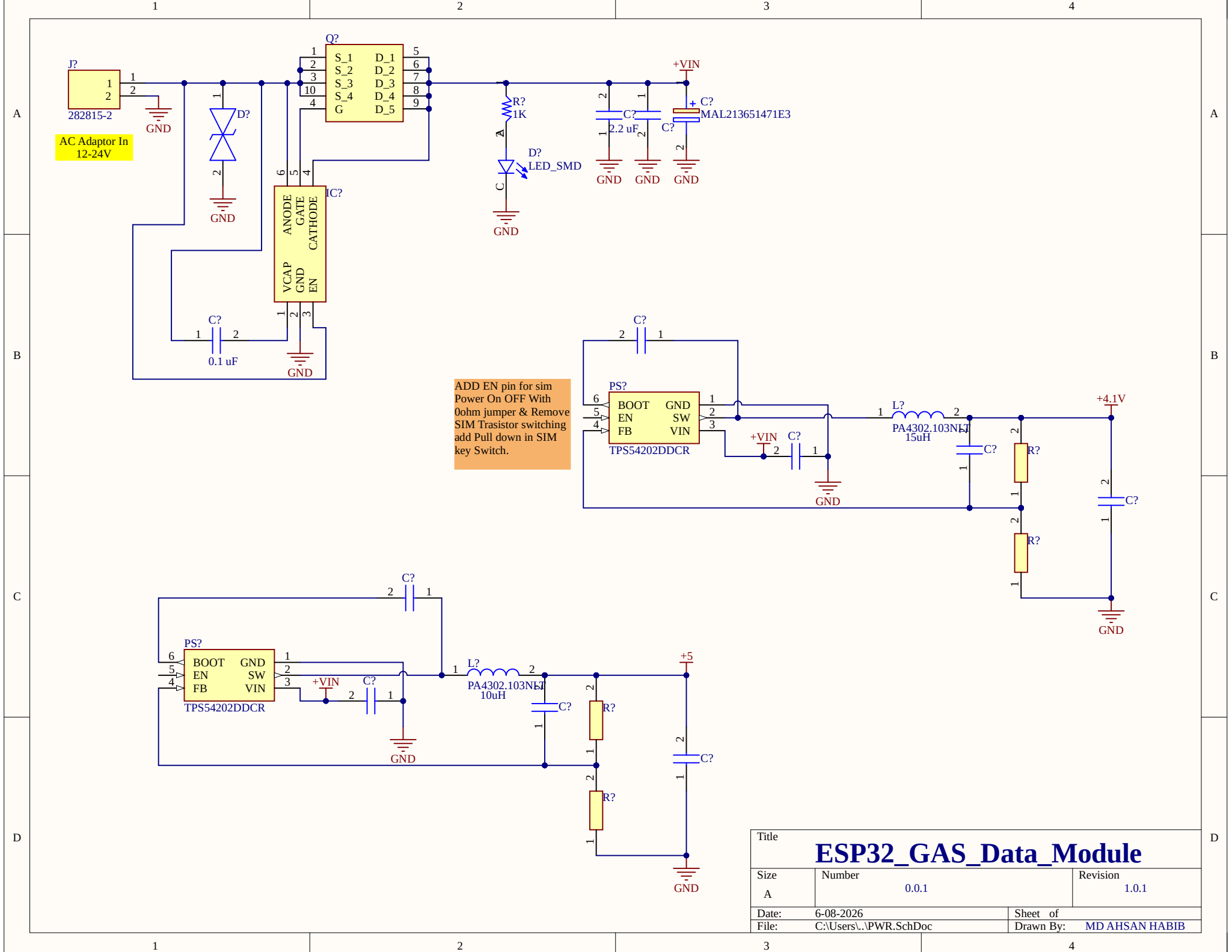


|   |  |   |  |   |  |   |  |
|---|--|---|--|---|--|---|--|
| 1 |  | 2 |  | 3 |  | 4 |  |
| A |  |   |  |   |  |   |  |
| B |  |   |  |   |  |   |  |
| C |  |   |  |   |  |   |  |
| D |  |   |  |   |  |   |  |
| 1 |  | 2 |  | 3 |  | 4 |  |

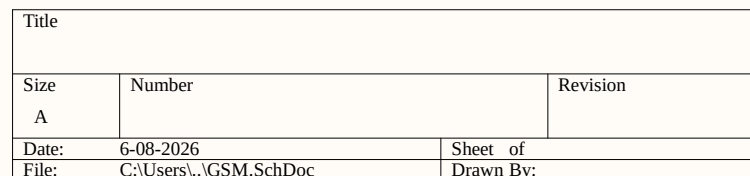
RO  
RE  
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DI

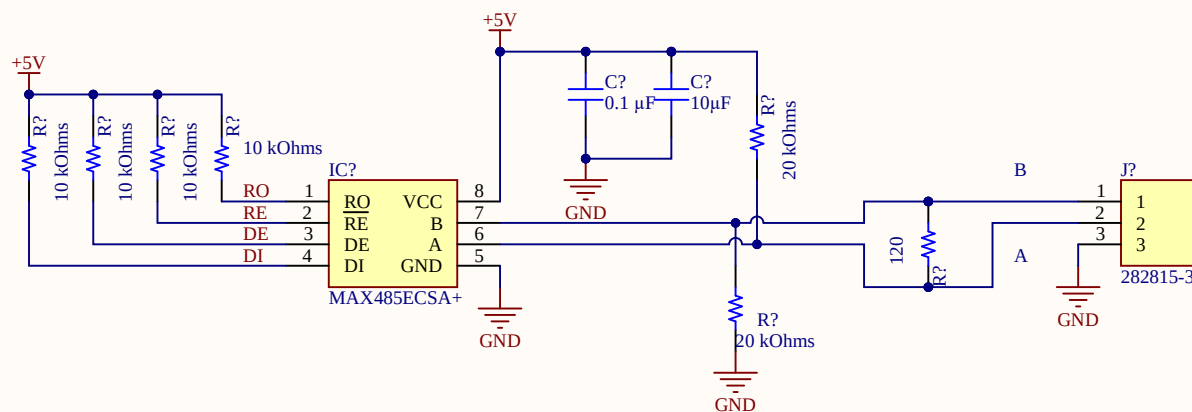
SIM\_EN  
SIM\_TX  
SIM\_RX

|           |                         |           |          |
|-----------|-------------------------|-----------|----------|
| Title     |                         |           |          |
| Size<br>A | Number                  |           | Revision |
| Date:     | 6-08-2026               | Sheet     | of       |
| File:     | C:\Users\...\MCU.SchDoc | Drawn By: |          |

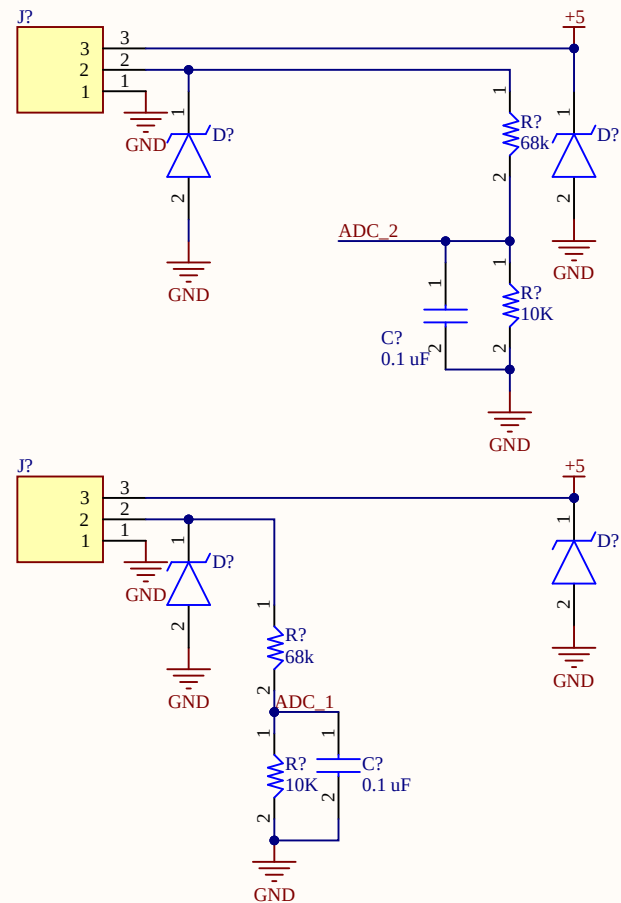


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|-------|-------------------------|-----------|-----------------------|--|
| Title |                         |           | ESP32_GAS_Data_Module |  |
| Size  | Number                  | Revision  |                       |  |
| A     | 0.0.1                   | 1.0.1     |                       |  |
| Date: | 6-08-2026               | Sheet     | of                    |  |
| File: | C:\Users\...\PWR.SchDoc | Drawn By: | MD AHSAN HABIB        |  |





| Title |                           |           |
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| A     |                           |           |
| Date: | 6-08-2026                 | Sheet of  |
| File: | C:\Users\...\RS485.SchDoc | Drawn By: |



| Title |                         |           |
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| Size  | Number                  | Revision  |
| A     |                         |           |
| Date: | 6-08-2026               | Sheet of  |
| File: | C:\Users\...\ADC.SchDoc | Drawn By: |

| Comment              | Description                                                          | Designator | Footprint                              | LibRef               | Quantity |
|----------------------|----------------------------------------------------------------------|------------|----------------------------------------|----------------------|----------|
| 0.1 µF               | CAP CER0.1UF 50V X7R0603                                             | C?         | CAP-0603_1608                          | C0603C104K5RAC7411   | 1        |
| 0.1 uF               | Capacitor                                                            | C?         | CAP0603                                | CC0603KFX7F9BB104    | 3        |
| 2.2 uF               | Capacitor                                                            | C?         | CAP0603                                | CC0603KFX7F9BB104    | 3        |
| 10µF                 | CAP CER 10UF 50V XSR1206                                             | C?         | CAP 1206_3216 - 1.8MM                  | 12065D106KAT2A       | 1        |
| 470 uF               | 220F 25V Aluminum Electrolytic Capacitors Radial, Can 2000 Hrs @105C | C?         | CAPPRD350W60D800 H1150                 | EEUFC1E221           | 2        |
| C0805C104M5RAC-TU    | Capacitor                                                            | C?         | CAPC2012X88N                           | C0805C104M5RAC-TU    | 2        |
| C5750XSRH106K230K A  | Capacitor                                                            | C?         | C5750_Commercial                       | C5750XSRH106K230K A  | 3        |
| CGA1A2CG1EB20J030 BA | Capacitor                                                            | C?         | CAPC0603X33N                           | CGA1A2CG1EB20J030 BA | 1        |
| GRM32ER61C476KE15 L  | Capacitor                                                            | C?         | CAPC3225X270N                          | GRM32ER61C476KE15 L  | 2        |
| GRM0335C1E560JA01 D  | Capacitor                                                            | C?         | GRM033D70J224ME01 D                    | GRM0335C1E560JA01 D  | 1        |
| MAL213651471E3       | Capacitor Polarised                                                  | C?         | CAPPRD500W60D1300 H2700                | MAL213651471E3       | 1        |
| LED_SMD              | Red 633nm LED Indication - Discrete 2.1V 0805 (2012 Metric)          | D?         | LEDSC125X200X90-2N                     | 5990110007F          | 2        |
| LED_TH               | Red 617nm LED Indication - Discrete 2V Radial                        | D?         | LED_WP7113ID                           | WP7113ID             | 1        |
| SMBJ5.0A             | TVSDiode (Uni-directional)                                           | D?         | SMBJ12A                                | SMBJ5.0A             | 4        |
| SMBJ24CA-13-F        | TVSDiode (Bi-directional)                                            | D?         | DIONM5336X250N                         | SMBJ24CA-13-F        | 1        |
| LM74700QDBVTQ1       | Integrated Circuit                                                   | IC?        | SOT95P280X145-6N                       | LM74700-Q1MKX-1_NOPB | 1        |
| MAX485ECSA+          | Integrated Circuit                                                   | IC?        | SOIC127P600X175-8N                     | MAX485ECSA+          | 1        |
| MMBT2222ALT1G        | Integrated Circuit                                                   | IC?        | SOT96P237X111-3N                       | MMBT2222ALT1G        | 1        |
| 282815-2             | Connector                                                            | J?         | SHDFFRA2V102FOX508_1X2_1016X1200X850 P | 282815-2             | 1        |
| 282815-3             | Connector                                                            | J?         | 2828153                                | 282815-3             | 3        |
| CONUFL001-SMD        |                                                                      | J?         | LINUX_CONUFL001-SMD                    | CONUFL001-SMD        | 1        |
| SIM8060-6-1-14-00-A  | 6 Position Card Connector NANO SM Surface Mount, Right Angle Gold    | J?         | GCT_SIM8060-6-1-14-00-A                | SIM8060-6-1-14-00-A  | 1        |
| SMD_SMD              | Connector Header Through Hole 4 position 0.098 (2.50mm)              | J?         | SMD_SMD                                | SMD_SMD              | 1        |
| PA4302.103NLT        | Inductor                                                             | L?         | PA4302103NLT                           | PA4302.103NLT        | 2        |
| TPS54202DDCR         | Power Supply                                                         | PS?        | SOT95P280X110-6N                       | TPS54202DDCR         | 2        |
| BSC014N06NSATMA1     | MOSFET (N-Channel)                                                   | Q?         | BSC010N04LS6ATMA1                      | BSC014N06NSATMA1     | 1        |
| 1K                   | Resistor                                                             | R?         | RES0603                                | RES                  | 3        |
| 4.7K                 | Resistor                                                             | R?         | RES0603                                | RES                  | 1        |
| 10 kOhms             | RES SMD 10K OHM 1% 1/10W 0603                                        | R?         | RES0603_1608                           | RT0603FRE0710KL      | 4        |
| 10K                  | Resistor                                                             | R?         | RES0603                                | RES                  | 2        |
| 20 kOhms             | RES SMD 10K OHM 1% 1/10W 0603                                        | R?         | RES0603_1608                           | RT0603FRE0710KL      | 2        |
| 47K                  | Resistor                                                             | R?         | RES0603                                | RES                  | 1        |
| 68k                  | Resistor                                                             | R?         | RES0603                                | RES                  | 2        |
| 120                  | RES SMD 10K OHM 1% 1/10W 0603                                        | R?         | RES0603_1608                           | RT0603FRE0710KL      | 1        |
| CROW040213K7FKEDHP   | Resistor                                                             | R?         | RESC1005X40N                           | CROW040213K7FKEDHP   | 1        |
| CROW040216K9FKEDHP   | Resistor                                                             | R?         | RESC1005X40N                           | CROW040216K9FKEDHP   | 1        |
| CROW0402100KFKEDHP   | Resistor                                                             | R?         | RESC1005X40N                           | CROW0402100KFKEDHP   | 2        |
| NC                   | Resistor                                                             | R?         | RES0603                                | RES                  | 1        |
| SIM800C              | Module: GSM/Bluetooth; GPRS, 2G, 1800MHz, 900MHz, 85.6kbps; SMD      | U?         | LCCL110P1570X1760X250-42N              | SIM800C              | 1        |